

AUDIO ADCs

The ADAU1701 has two Σ - Δ ADCs. The signal-to-noise ratio (SNR) of the ADCs is 100 dB, and the THD + N is -83 dB.

The stereo audio ADCs are current input; therefore, a voltage-to-current resistor is required on the inputs. This means that the voltage level of the input signals to the system can be set to any level; only the input resistors need to be scaled to provide the proper full-scale current input. The ADC0 and ADC1 input pins, as well as ADC_RES, have an internal 2 k Ω resistor for ESD protection. The voltage seen directly on the ADC input pins is the 1.5 V common mode.

The external resistor connected to ADC_RES sets the full-scale current input of the ADCs. The full range of the ADC inputs is 100 μ A rms with an external 18 k Ω resistor on ADC_RES (20 k Ω total, because it is in series with the internal 2 k Ω). The only reason to change the ADC_RES resistor is if a sampling rate other than 48 kHz is used.

The voltage-to-current resistors connected to ADC0/ADC1 set the full-scale voltage input of the ADCs. With a full-scale current input of 100 μ A rms, a 2.0 V rms signal with an external 18 k Ω resistor (in series with the 2 k Ω internal resistor) results in an input using the full range of the ADC. The matching of these resistors to the ADC_RES resistor is important to the operation of the ADCs. For these three resistors, a 1% tolerance is recommended.

Either the ADC0 and/or ADC1 input pins can be left unconnected if that channel of the ADC is unused.

These calculations of resistor values assume a 48 kHz sample rate. The recommended input and current setting resistors scale linearly with the sample rate because the ADCs have a switched-capacitor input. The total value (2 k Ω internal plus external resistor) of the ADC_RES resistor with sample rate f_{S_NEW} can be calculated as follows:

$$R_{total} = 20 \text{ k}\Omega \times \frac{48,000}{f_{S_NEW}}$$

The values of the resistors (internal plus external) in series with the ADC0 and ADC1 pins can be calculated as follows:

$$R_{Input \text{ Total}} = (rms \text{ Input Voltage}) \times 10 \text{ k}\Omega \times \frac{48,000}{f_{S_NEW}}$$

Table 13 lists the external and total resistor values for common signal input levels at a 48 kHz sampling rate. A full-scale rms input voltage of 0.9 V is shown in the table because a full-scale signal at this input level is equal to a full-scale output on the DACs.

Table 13. ADC Input Resistor Values

Full-Scale RMS Input Voltage (V)	ADC_RES Value (k Ω)	ADC0/ADC1 Resistor Value (k Ω)	Total ADC0/ADC1 Input Resistance (External + Internal) (k Ω)
0.9	18	7	9
1.0	18	8	10
2.0	18	18	20

Figure 17 shows a typical configuration of the ADC inputs for a 2.0 V rms input signal for a f_s of 48 kHz. The 47 μ F capacitors are used to ac-couple the signals so that the inputs are biased at 1.5 V.

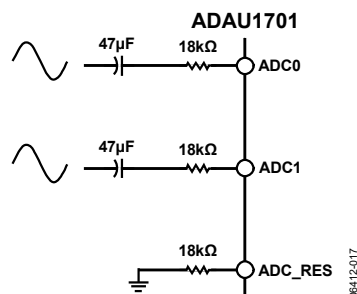


Figure 17. Audio ADC Input Configuration